DFM analysis report_JLCDFM

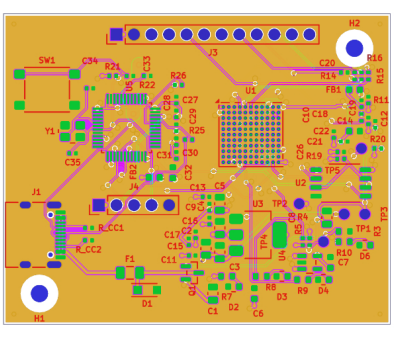
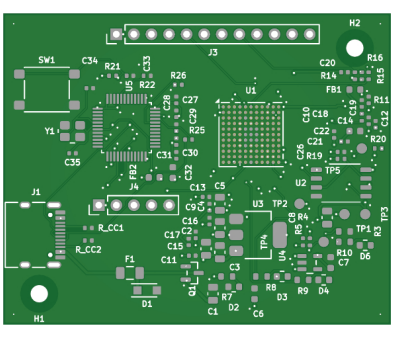
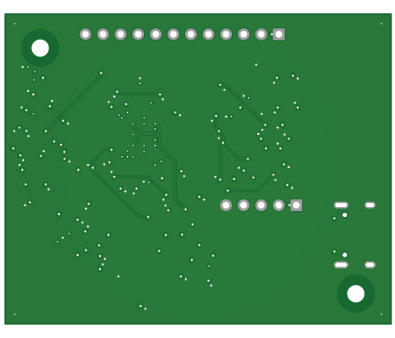
File name：**ice40_bga_node_gerbers.zip**

Report generated at：**2026-11-06 14:42:27**

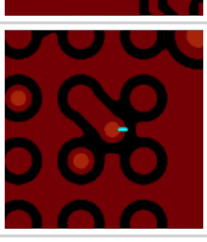
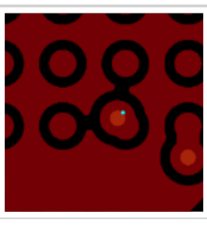
PCB layers：**6**

PCB size：**5.6x4.5cm**

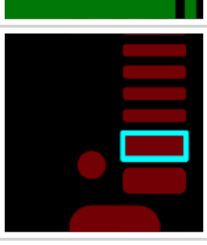
Analyze project：☑ **PCB DFM** ☐ **SMT DFM**

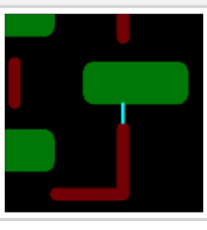
PCB DFM>Routing layer analysis

Analyze project	Analysis results	PCB screenshot	Layer distribution	Statistics
Sharp trace corner (Check for sharp corners in traces)	No	No	No	Danger：0 Warning：0 Good：0
BGA pad (Check BGA pads on the board)	null Warning		ice40_bga_node-F_Cu.gtl	Danger：0 Warning：50 Good：0
Via placed within a pad (Check if there is a via placed within a pad)	No	No	No	Danger：0 Warning：0 Good：0
Trace to board edge (Detect traces too close to the board edge)	No	No	No	Danger：0 Warning：0 Good：0
Trace spacing (Measure spacing between adjacent parallel traces)	No	No	No	Danger：0 Warning：0 Good：0
Unconnected trace end (Free-standing trace ends not connected to pads)	No	No	No	Danger：0 Warning：0 Good：0
Trace width (Trace width information)	0.13mm Warning		ice40_bga_node-F_Cu.gtl ice40_bga_node-B_Cu.gbl	Danger：0 Warning：22 Good：50
Fiducial (Detect fiducial marks on the board)	No	No	No	Danger：0 Warning：0 Good：0
Pad to board edge (Measure distance of pads from the board edge)	No	No	No	Danger：0 Warning：0 Good：0
Pad spacing (Measure pad to pad spacing)	0.1mm Danger		ice40_bga_node-F_Cu.gtl ice40_bga_node-B_Cu.gbl	Danger：1 Warning：2 Good：48
Plated through-hole to trace clearance (Measure clearance of plated through-holes to traces)	0.24mm Good		ice40_bga_node-F_Cu.gtl ice40_bga_node-B_Cu.gbl	Danger：0 Warning：0 Good：13
Annular ring (Annular ring width of pads compared to holes)	0.08mm Danger		ice40_bga_node-F_Cu.gtl ice40_bga_node-B_Cu.gbl	Danger：19 Warning：81 Good：0
tht to smd (Detect clearance of vias to pads)	2.13mm Warning		ice40_bga_node-F_Cu.gtl ice40_bga_node-B_Cu.gbl	Danger：0 Warning：2 Good：49
Via to pad (Detect clearance of vias to pads)	0mm Danger		ice40_bga_node-F_Cu.gtl	Danger：3 Warning：2 Good：1

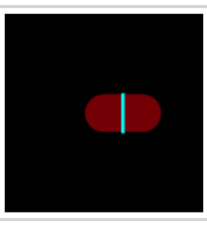
PCB DFM>Soldermask layer analysis

Analyze project	Analysis results	PCB screenshot	Layer distribution	Statistics
Soldermask bridge (Detect distance between parallel soldermask opening edges)	0.2mm Good		ice40_bga_node-F_Mask.gts	Danger：0 Warning：0 Good：50
Solder mask opening exposing trace (Detect clearance of solder mask openings to nearby traces)	0.1mm Warning		ice40_bga_node-F_Mask.gts	Danger：0 Warning：50 Good：0
Soldermask opening with multiple segments (Check if solder mask openings are constructed from multiple geometric shapes)	null Warning		ice40_bga_node-F_Mask.gts	Danger：0 Warning：4 Good：0
Negative soldermask expansion (Detect solder mask openings smaller than their corresponding pads)	0.04mm Warning		ice40_bga_node-F_Mask.gts	Danger：0 Warning：30 Good：0

PCB DFM>Silkscreen layer analysis

Analyze project	Analysis results	PCB screenshot	Layer distribution	Statistics
Silkscreen to pad (Detect clearance of silkscreen to pads)	0.25mm Good		ice40_bga_node-F_Silkscreen.gto	Danger：0 Warning：0 Good：6
Silkscreen to hole (Detect clearance of silkscreen to holes)	No	No	No	Danger：0 Warning：0 Good：0
Silkscreen line width (Check silkscreen line width)	0.18mm Good		ice40_bga_node-F_Silkscreen.gto	Danger：0 Warning：0 Good：50

PCB DFM>Drill layer analysis

Analyze project	Analysis results	PCB screenshot	Layer distribution	Statistics
Unconnected via (Detect isolated unconnected vias)	No	No	No	Danger：0 Warning：0 Good：0
Missing plated through-hole (Detect top and bottom pads at the same location without plated through-holes)	No	No	No	Danger：0 Warning：0 Good：0
Unconnected via (Detect isolated unconnected vias)	No	No	No	Danger：0 Warning：0 Good：0
Plated through-hole spacing (Measure spacing between plated through-holes)	No	No	No	Danger：0 Warning：0 Good：0
Short slot detection (Detect slots shorter than twice their width)	No	No	No	Danger：0 Warning：0 Good：0
Slot width check (Measure slot width)	0.6mm Danger		ice40_bga_node.drl	Danger：4 Warning：0 Good：0
Via to PTH spacing (Measure spacing of vias to plated through-holes)	No	No	No	Danger：0 Warning：0 Good：0
Unconnected via (Detect isolated unconnected vias)	No	No	No	Danger：0 Warning：0 Good：0